$$\begin{cases} \dfrac{dy}{dx} = f(x,y) \\ x=0 \quad y(0) = y_{inicial} \end{cases}$$

$(0, y_{inicial})$

M. EULER

$$\frac{dy}{dx} \approx \frac{\Delta y}{\Delta x}$$

$$\frac{dy}{dx} = \frac{\Delta y}{\Delta x} + \epsilon$$

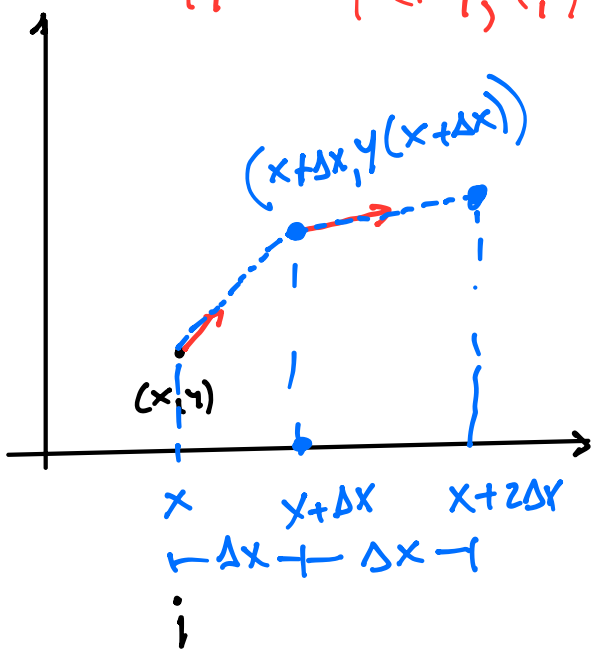
$$\Delta y = f(x,y)\,\Delta x$$

$$\Delta y = y(x+\Delta x) - y(x)$$

$$y(x+\Delta x) - y(x) = f(x,y)\,\Delta x$$

$$\boxed{y(x+\Delta x) = y(x) + f(x,y)\,\Delta x} + \epsilon$$

$$y_{i+1} = y_i + f(x_i, y_i)\,\Delta x$$



$x_i \to x$

$x_{i+1} \to x + \Delta x$

$y_i \to y(x_i)$

$y_{i+1} \to y(x+\Delta x)$

(x_0, y_0)

$$y_1 = y_0 + f(x_0, y_0)\,\Delta x \longrightarrow (\overset{x_0+\Delta x}{x_1}, y_1)$$

paso

$$y_2 = y_1 + f(x_1, y_1)\,\Delta x \longrightarrow (\overset{x_0+2\Delta x}{x_2}, y_2)$$

$$y_3 = y_2 + f(x_2, y_2)\Delta x \rightarrow (\overset{x_0+3\Delta x}{x_3}, y_3)$$

x	y
x_0	y_0
x_1	y_1
x_2	y_2
$\vdots$	$\vdots$
x_N	y_N

Δx

modelo

$$m\frac{dv}{dt} = -mg - \mu v \qquad \frac{dy}{dx} = f(x,y)$$

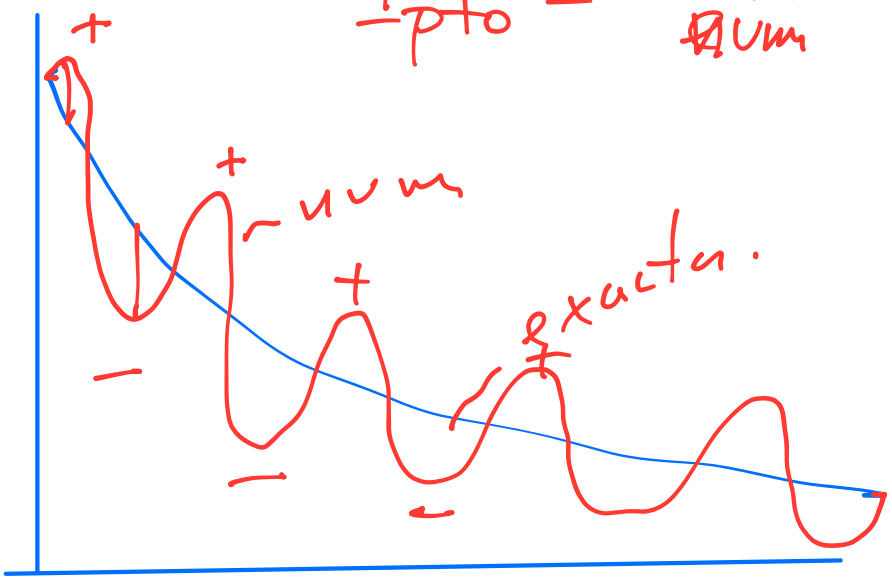
$v(t) \quad t=0 \quad v(0) = v_{inicial}$

$(0, v_{inicial})$

$$\frac{dv}{dt} = -g - \frac{\mu}{m}v = f(v,t)$$

$$\int \frac{dv}{-g - \frac{\mu}{m}v} = \int dt$$

$$E_{pto} = \text{Valor num} - \text{Valor exact.}$$



$$E_{TOT} = \sum_{i=1}^{N} |E_i| \qquad \text{norma } L_\infty$$

$$E_{TOT} = \frac{\sum_{i=1}^{N} E_i^2}{N}$$

$$E_{TOT} \leq \underline{\underline{TOL}} = 1 \cdot 10^{-6}$$

$$1 \cdot 10^{-3}$$

$$m\frac{d^2x}{dt^2} = -mg - \mu v \qquad m\frac{dv}{dt} = -mg - \mu v$$

$$m\frac{d^2x}{dt^2} = -mg - \mu\frac{dx}{dt}$$

1) $\frac{dx}{dt} = v$

$x(t)$

$v(t)$

2) $m\frac{dv}{dt} = -mg - \mu v$

<u>notación vectorial</u>

$$\vec{y} = \begin{pmatrix} x \\ v \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \end{pmatrix}$$

1) $\frac{dy_0}{dt} = y_1$

2) $$m\frac{dy_1}{dt} = -mg - \mu y_1$$

$$\frac{dy_0}{dt} = y_1$$

$$\frac{dy_1}{dt} = -g - \frac{\mu}{m} y_1$$

$$\begin{pmatrix} \frac{dy_0}{dt} \\ \frac{dy_1}{dt} \end{pmatrix} = \begin{pmatrix} y_1 \\ -g - \frac{\mu}{m} y_1 \end{pmatrix}$$

$$\frac{d}{dt}\vec{y} = \begin{pmatrix} y_1 \\ -g - \frac{\mu}{m} y_1 \end{pmatrix} = \vec{F}(t, \vec{y})$$

$$x''' - 3x'' + x' - \sin t = 0$$

$$u = x'$$

$$z = u' = x''$$

$$z' - 3z + u - \sin t = 0$$

$$\begin{cases} x' = u \\ u' = z \\ z' = \sin t + 3z - u \end{cases}$$

$$\vec{p} = \begin{pmatrix} x \\ u \\ z \end{pmatrix} = \begin{pmatrix} p_0 \\ p_1 \\ p_2 \end{pmatrix}$$

$$P_0' = P_1$$

$$P_1' = P_2$$

$$P_2' = \sin t + 3P_2 - P_1$$

$$\begin{pmatrix} P_0 \\ P_1 \\ P_2 \end{pmatrix}' = \begin{pmatrix} P_1 \\ P_2 \\ \sin t + 3P_2 - P_1 \end{pmatrix}$$

$$\frac{d\vec{y}}{dt} = \vec{F}(t, \vec{y})$$

$$\vec{y}_{i+1} = \vec{y}_i + \Delta t \, \vec{F}(t_i, \vec{y}_i)$$

$$\begin{pmatrix} \frac{dy_0}{dt} \\ \frac{dy_1}{dt} \end{pmatrix} = \begin{pmatrix} y_1 \\ -g - \frac{\mu}{m} y_1 \end{pmatrix} \qquad \vec{y} = \begin{pmatrix} x \\ v \end{pmatrix}$$

$$\begin{cases} \frac{d}{dt}\vec{y} = \begin{pmatrix} y_1 \\ -g - \frac{\mu}{m} y_1 \end{pmatrix} = \vec{F}(t, \vec{y}) \\ t = 0 \quad \vec{y}_0 = \begin{pmatrix} x_{ini} \\ v_{ini} \end{pmatrix} \end{cases}$$

t	y_0 x	y_1 v
t_0	x_0	v_0
t_1	x_1	v_1
t_2	x_2	v_2

$\vec{y}$

t	t_0	t_1	t_2 ...
X	X_0	X_1	X_2 ...
V	V_0	V_1	V_2 ...